

Adding a reusable feature channel beside a prediction service you
cannot change:
an evaluated release problem, one confirmed effect, and four
negative method results

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Study 4 integrated from evidence commit `69e790af`; a fifth programme is registered as pending and carries no result.

Abstract

An organisation already publishes a prediction service—fixed probability vectors that downstream recipients consume and cannot be asked to re-accept. We formalise and evaluate what happens when it adds a *second*, reusable feature channel for one of those recipients without altering a single published number, and tries to limit what that recipient, and that recipient colluding with another, can *newly* infer about sex and race. The contribution is an implemented release interface with a measurement contract—per-recipient views, a coalition view, independently fitted attackers, and incremental disclosure always reported beside absolute disclosure—together with what three completed studies of one mechanism family measured through it. Four findings. (i) Exact preservation of the published outputs is structural and holds bitwise on all 210 released views; it does not imply stable service *accuracy*, which moves by up to 0.015 nats across one year. (ii) On a locked, previously unused survey year, a penalty built from the coalition’s joint view beats both an equal-strength and an equal-total-mass local control under simultaneous 95% intervals: 14 of 16 sensitive cells strictly better, none worse, two unresolved. The paired residence constraint is a *point* rule, and interval non-inferiority at the same margin holds in 0 of 4 comparisons. (iii) Two successive refinements of the mechanism—a penalty nonlinear in the channel, then an exactly rotation-invariant repair of that penalty—both fail. The repair is verified exact to 10^{-15} , and the release it produces discloses significantly *more* than the defective objective it repairs on 8 of 48 sensitive cells, with none significantly less. (iv) The first executed external-method comparisons put adapted LEACE, SPLINCE and two OptNet-ARL policies below the developed mechanism on every sensitive endpoint they resolve—31 of 32 arm $\times$ endpoint $\times$ weighting cells; adapted LEACE also costs 0.0098 nats of authorised residence capability against the channel it transformed, a significant loss. (v) A fourth study refines the channel by direct adversarial training against a refreshed attacker ensemble, and fails on both years: across its frontier comparison family 189 cells are significantly worse and 18 significantly better, *every* one of the 18 on residence utility and none on a sensitive endpoint. Its one positive—that coalition conditioning beats a local control—holds robustly only against the *weak* local control; against the strength-matched control it survives at the family-adjusted level on one endpoint at one width and does not survive the simultaneous correction its negative results are held to. Lower recovery together with lower retained capability is a trade-off, not dominance. We therefore report an empirical and problem-formulation contribution and four negative method results; we list every claim of our own that this revision withdraws or narrows; and we review a fifth, two-track programme *before* any of its outcomes exist, with its cells left empty.

1 What the paper does

Suppose a company already publishes two prediction services: a credit-style model that returns probabilities to one business partner, and a coverage model that returns probabilities to another. Both partners have built systems on those exact numbers, so the company cannot revise them. It now wants to give the first partner something more useful—a compact reusable feature vector built from what the published predictions leave

unexplained—while making sure that the partner, and the two partners pooling what they hold, cannot use it to work out an applicant’s sex or race any better than the published predictions already allow. That is the problem this paper implements and measures.

Contribution, precisely. We build the interface and the measurement contract, and we report what a bounded programme of experiments measured through them. A competitive mechanism remains a hypothesis this project is still testing; it is not something this paper claims, and it is not something this paper has abandoned.

- **The interface.** A mechanism appends a channel Z to one recipient’s published output and modifies nothing. Preservation is then an *identity*, not a bounded perturbation, and is checked bitwise on every released view.
- **The comparisons it enables.** Because Z is separable, heterogeneous mechanisms become directly comparable at matched width on the same wire: spectral moment penalties, an adversarially trained neural channel, and closed-form linear erasers, all evaluated by the same independently fitted attacker slate against the same recipient-specific and coalition-specific audit roles, with H -only recovery always subtracted and always also shown unsubtracted.
- **The empirical finding.** On a sealed year, a penalty built from the coalition’s joint view lowers measured recovery relative to two local-only controls matched on penalty strength and on total penalty mass. Everything the project then developed to close the remaining gap to the strongest comparator failed, and three adapted published methods beat it.
- **What the fourth study adds.** Direct adversarial refinement of the channel does not improve the frontier (§9). Two by-products are worth more than the headline: a closed-form eraser applied to a channel this project already had is now confirmed on a *second* year, because fifteen previously missing baseline transports were completed and each was admitted only after rebuilding its 2018 release bitwise; and the study’s own scope correction shows that an unmatched attack opportunity had been crediting an *identical* channel with 0.0155 nats of apparent protection. That is an evaluation correction, not a gain.
- **What is pending.** A fifth programme—a factorial neural experiment and a coalition-conditioned rank-reducing projection—is reviewed here for mathematical validity *before* it runs (§10). No outcome from it has been read, no protocol for it has been committed, and this paper prints no forecast in its place.

As of this revision this is an empirical and problem-formulation paper, not a method paper. We do not present a deployable mechanism, and we do not yet have one. Method development continues; §10 states exactly what a candidate would have to show to be described otherwise. We are not offering new mathematics: known concept erasure, eigenvalue-sign rank selection, minimax adversarial training and conditional-moment characterisations are prior work, cited as such. Adversarially protected representations that transfer to new tasks are not new either [Madras et al., 2018]. What we did not find, after a bounded search, is a prior *evaluation* of recipient-specific and combined-access policies over an already-published output that the mechanism may not touch (§4); that is absence of evidence after a bounded search, not a priority claim.

2 The interface, and four distinctions we hold to throughout

Each person has covariates X , sensitive attributes S (sex, 2 classes; race, 9 classes) and task labels. Two services are frozen and published before anything here happens: $H_A(X) \in [0, 1]^4$, the full income-over-\$50k and employment probability vectors released to recipient A, and $H_B(X) \in [0, 1]^2$, the coverage vector released to recipient B. A mechanism may add $Z \in \mathbb{R}^{16}$ for A *only*. So A sees (H_A, Z) , B sees H_B , and the coalition AB sees (H_A, Z, H_B) . H without a subscript means the no-channel interface. The policy authorises A for income, employment and residence and B for coverage and commute, and forbids sex and race everywhere

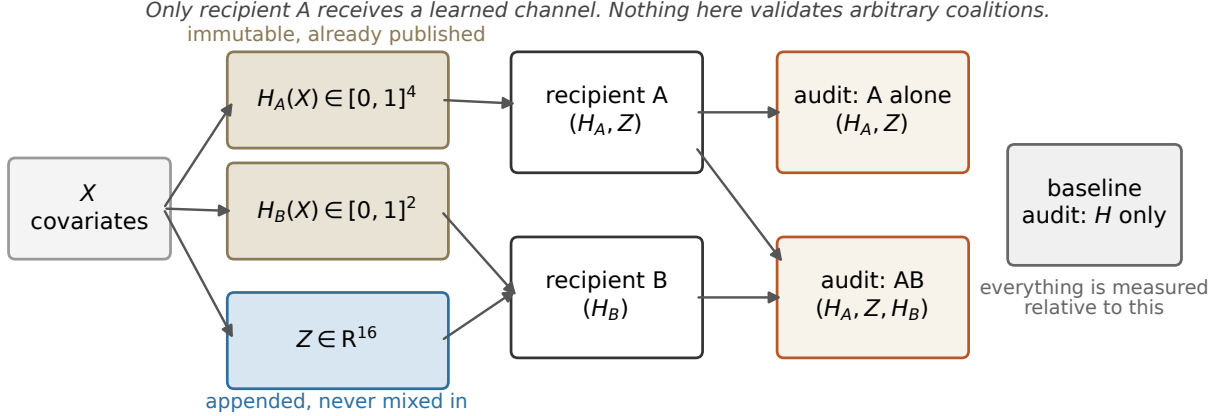


Figure 1: The release graph. H_A and H_B are already published and are never modified; Z is appended to A ’s wire alone. Three audit views are scored: A alone, the coalition AB , and the H -only baseline that every number in this paper is measured against. This is a schematic of the interface, not a result.

plus nine cross-role combinations—eleven forbidden roles in all. The adversary holds labelled data from the release year, fits fixed attack families on its own pool, selects on a validation pool, and sees only its legal view.

Three names recur. J is a 16-dimensional adversarially trained channel recovered from earlier work and never retrained; it is the strongest comparator in the study and it is *internal*. $L1$ and $L2$ are local controls whose penalty uses only A ’s own view, matched to the candidate on penalty strength and on total penalty mass respectively, so that a coalition effect cannot be confused with “more penalty”. $C1$ adds a penalty term built from the coalition’s joint view; it is the candidate.

Four distinctions carry the argument and are never merged.

Preservation is architectural; accuracy is empirical. Z is concatenated to H_A and never mixed into it, so H is preserved by construction—a property of concatenation, not a theorem. It does *not* follow that the service stays accurate: on identical released vectors, employment log loss rises by 0.0138 nats and income falls by 0.0145 between the two years (Appendix C).

Added capability is measured against the correct H -only view. A channel “adds capability” only if independently fitted probes do better with (H_A, Z) than with H_A .

Added disclosure is a finite-auditor log-loss gain, not an information bound. Every recovery number is the improvement a validation-selected attacker achieves over the same attacker family restricted to H . It is a floor on what is recoverable and never a ceiling (§11).

Recipients have different permitted tasks. AB ’s authorised union is not either singleton’s permission set, and the audit roles differ accordingly.

Two scope limits bound every claim below. Only recipient A receives a channel: nothing here demonstrates joint optimisation of channels for several recipients, and the coalition penalty is a constraint on one channel given another recipient’s fixed view, not a multi-channel design. And although residence and commute labels are excluded from every representation fit—the sense in which residence is held out—residence outcomes have been inspected repeatedly across this line of work, so residence is not a pristine unseen task and we do not claim unseen-task generality.

3 The mechanism family, and what is a theorem here

Let T be a frozen PCA32 teacher. Build $\phi(T)$ from standardised T and 96 random Fourier features [Rahimi and Recht, 2007]; residualise against a degree-2 polynomial basis in the released service probabilities and

eigen-whiten to V with $V^\top V/n = I$. Utility is $U = C_{VR}C_{RV}$ for the residualised teacher R : for $W^\top W = I$, $\text{tr}(W^\top U W)$ is exactly the training reduction in least-squares reconstruction error of R . Protection matrices P_j are built from household-grouped 3-fold cross-fitted nuisances $m_j(H)$ [Chernozhukov et al., 2016] and the residuals $\text{onehot}(S_j) - m_j(H)$, aggregated over local roles, coalition roles and a marginal variant. The released map is the top-16 eigenvectors of $U_{\text{norm}} - \lambda P$. Inference needs only T and H_A ; H_B enters fitting and auditing and is never an inference input.

Study 2 replaced the fixed penalty matrix by one computed from degree-two monomials of the channel output, so the objective is no longer $\text{tr}(W^\top A W)$ for any W -independent A , and added a predeclared rank-8 variant. Study 3 replaced that penalty again with two blocks that are exactly invariant under $W \mapsto WQ$ for orthogonal Q : a Frobenius-weighted quadratic moment block, for which rotation acts as $M \mapsto Q^\top M Q$, and an exact average of three Gaussian kernels on a frozen 512-row subset, which sees the data only through pairwise distances.

What is proved, and over what. Four statements are elementary and we prove them in Appendix A; nothing else in this paper is a theorem.

1. Appending Z without modifying H preserves H exactly. This is a property of concatenation.
2. An unrestricted recipient holding $[H, Z]$ can ignore Z , so its optimal population log loss is no worse than with H alone. This bounds the *population optimum*, not any fitted model, and our measured increments are sometimes negative.
3. Under unrestricted Bayes prediction and ordinary log loss, the population risk difference between H and (H, Z) equals $I(S; Z | H)$. Our validation-selected finite attackers do not attain those infima, so the reported differences bound that quantity in neither direction, and mutual-information estimation has its own formal limits [McAllester and Stratos, 2020].
4. For the repaired quadratic block, orthogonal reparameterisation preserves the Frobenius norm, and an exact radial kernel is invariant to the corresponding distance-preserving rotations. Both hold identically, subject to stated finite-precision conditions; measured, the objective moves by at most 5.6×10^{-17} under random rotations.

Ky Fan’s maximum principle [Fan, 1949] gives a restricted global optimum for the *original* fixed-matrix family only. The refinements have no fixed matrix to take a spectrum of, so they inherit no optimality statement whatever, and a suboptimal solution remains a live explanation for their results. Eigenvalue-sign rank selection is SARL’s Theorem 3, OptNet-ARL’s Theorem 4.1 and K-TOpt’s Corollary 4.1 [Sadeghi et al., 2019, 2021, 2022]; we fixed $r = 16$ to match the width of the neural comparator, which is a comparison choice, not an optimum. The moment condition is a finite fitted subset of the L^2 characterisation of conditional independence used by kernel CI tests [Zhang et al., 2011, Strobl et al., 2019]—Lemma 2(v) of Zhang et al. [2011], attributed there to Daudin—quantified over $f \in \{Z_\ell b_k(H)\}$ rather than over every square-integrable f . Vanishing fitted moments therefore establish no conditional independence, and a non-zero moment may reflect nuisance error: one of our own fixtures shows a misspecified $m(H)$ manufacturing a penalty more than ten times the oracle value under *exact* conditional independence.

4 Relation to prior work

The closed-form line of Sadeghi, Dehdashtian and Boddeti already solves this shape of problem. SARL [Sadeghi et al., 2019] takes eigenvectors of a utility-minus- λ -dependence matrix; OptNet-ARL [Sadeghi et al., 2021] keeps closed-form ridge players inside a learned encoder and states in one sentence that it generalises to several sensitive attributes; K-TOpt [Sadeghi et al., 2022] adds random Fourier features and evaluates on Folktables [Ding et al., 2021]; U-FaTE [Dehdashtian et al., 2024] adds a *conditional* penalty. Our marginal arms are, up to parametrisation, SARL’s construction on whitened residuals—verified by bitwise rebuild (§8). Adversarial fair *and transferable* representation learning is Madras et al. [Madras et al., 2018], and we claim nothing new for that combination. Linear concept erasure [Belrose et al., 2023, Holstege et al., 2025] supplies an exact guarantee against *linear* adversaries on a representation the releaser owns and may edit.

study	construction	evidence commit	status	what it established
1	locked ACS 2017 transport	349efa45	confirmatory	coalition effect confirmed; the .001 rule it did not establish
2	nonlinear / rank	c37807e4	development	negative; penalty was coordinate-dependent, diagnosed after evaluation
3	invariant repair + external baselines	73903b7f	development	negative; a closed-form eraser matched J and beat the developed mechanism
4	direct adversarial channel refinement	69e790af	development	negative ; nominee conjunction UNSUPPORTED on both years
5	factorial neural + coalition-conditioned projection	unregistered	PENDING	NO OUTCOME READ; protocol not yet committed

Table 1: Five studies, kept apart by what each is evidence *for*. Only Study 1 is confirmatory. Study 3 is a *new formulation designed to repair a defect Study 2 diagnosed after evaluation*; it does not retroactively repair Study 2’s evaluated maps, and its 2017 pass is development on a partition Study 1 already spent, not a second sealed-year confirmation. Study 4 is complete and integrated in §9. Study 5 has no outcome and no committed protocol. Only Study 1 is confirmatory. Studies 2–4 are development on pools this project has used repeatedly, and no computation on them can undo that. Study 5 is two tracks under review in this revision *before* any outcome exists: its slot registry is a plan, not an execution record, and this paper carries its cells as pending with no forecast. 2016 remains sealed and unscored.

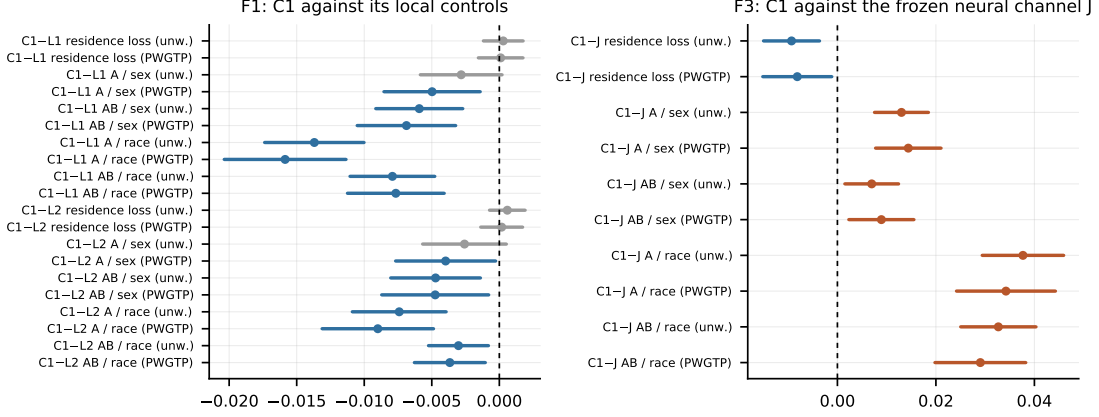
Adversarial removal is known to leave recoverable information [Elazar and Goldberg, 2018], and least-privilege learning has fundamental limits [Stadler et al., 2024]. **Reusing a known solver, a conditional-moment characterisation or a rank rule is attribution, not innovation.**

Three things differ, with their weight stated honestly. *What we condition on:* U-FaTE’s conditional measure is stratification over a discrete label, and H_A is a pair of continuous probability vectors, so we substitute cross-fitted nuisance residuals; H is also a *released artifact* rather than the label to be predicted. *What utility means:* reconstruction of a residualised teacher rather than dependence with a label. *Recipient structure:* separate local and coalition penalties over a fixed prior output, with only one recipient receiving a channel. The first two are adaptations of known components. For the third, the nearest candidate we found is weaker prior art than its title suggests: the multi-consumer information-theoretic line [Sankar et al., 2013] designs one sanitised release consumed by many users, and we could not confirm from its text that it carries a per-recipient release or a collusion term—we mark that entry ambiguous rather than assert it. Appendix B gives the full comparison table over five features (fixed pre-existing output, reusable extra features, recipient-specific permissions, coalition-aware training, independent attack evaluation), with every entry we could not verify from a full text marked as unverified rather than guessed.

5 Evidence status, and a chronology that keeps five studies apart

California ACS one-year PUMS, ages 19–34, three household splits throughout, with disjoint pools for representation fitting, source validation, task fitting and validation, attacker fitting and validation, and evaluation. Attacks are logistic regression, two restarted MLPs at 120/360 epochs, two gradient-boosted trees and a random-feature kernel-ridge family, plus every legal projection of H ’s own attacks and of singleton candidates into the coalition view. Uncertainty is a paired household-cluster bootstrap with single-step $\max\{-t\}$ simultaneous intervals inside each pre-declared family.

The distinction Table 1 enforces is worth stating in words, because it is easy to blur. Study 2 fitted and evaluated maps, and *afterwards* diagnosed that its penalty was coordinate-dependent. Study 3 wrote a new objective that is provably free of that defect, and refit under it. Those are three different things: a diagnosed bug, a new formulation, and a rerun. The Study 2 numbers in this paper remain numbers for the defective objective and are labelled so; the Study 3 numbers are numbers for a different construction, not corrected versions of Study 2’s.



paired difference (nats); left of zero favours the candidate. Blue: interval entirely below zero. Orange: entirely above. Grey: crosses zero (unresolved, not unaffected).

Figure 2: Paired differences with simultaneous 95% intervals inside each pre-declared family, fresh 2017 attacks, both weightings, candidate minus comparator in nats. **Left, F1:** $C1$ against its two local controls. The two grey A/sex rows are *unresolved*—the comparison is undecided, not the endpoint unaffected. **Right, F3:** $C1$ against J .

6 Study 1: the confirmed coalition effect, and the rule it did not establish

On the sealed 2017 partition, $C1$ has a simultaneous-interval advantage over both $L1$ and $L2$ under both weightings, and the registered **advantage** rule fired in all four contrast $\times$ weighting cells. The effect is real and not uniform: of the 16 sensitive cells in family F1, **14 have adjusted intervals entirely below zero, none is above zero, and two are unresolved** ($C1$ versus $L1$ on A/sex unweighted, $[-0.00585, +0.00020]$, and versus $L2$, $[-0.00568, +0.00052]$). Frozen 2018 attacks reproduce the ordering, so it is not an artifact of refitting attackers on the new year. Every augmented interface clears the 0.01-nat residence reference over H in every seed, weighting and mode.

The paired residence rule is a point criterion and only a point criterion. As implemented it is $\text{seed-mean}(C1 - \text{comparator}) \leq 0.001$ on residence log loss; the four values are 0.00029, 0.00010, 0.00059 and 0.00017, so it passes. It does *not* follow that the cost is below 0.001: **zero of four comparisons support interval non-inferiority at that margin**, missing it by 2% to 35% even under the most permissive procedure we computed, and zero support two-sided equivalence (Appendix C). “No statistically detected cost” and “a cost bounded below 0.001” are different statements and only the first is supported. The same applies to the harm side: the **advantage** rule requires that no sensitive endpoint be significantly worse, which is an absence-of-evidence condition, and on the two unresolved A/sex cells up to 5×10^{-4} nats of *extra* local sex recovery is not excluded.

And the mechanism is not competitive with J . $C1$ is significantly worse than J on all four sensitive endpoints under both weightings and on both source probes, and better only on residence, $-0.0093 [-0.0150, -0.0036]$. That ordering survives matching the two channels on average utility in both directions (Appendix C). It does *not* make J dominant: J carries materially less residence capability than every spectral arm, so neither channel Pareto-dominates the other, and J is an internal channel, so this is evidence about J and not about the state of the art.

7 Studies 2 and 3: two refinements, both negative

Study 2 asked whether that gap closes if the penalty acts on nonlinear functions of the channel, or the released rank is reduced, or both. Both factors moved measured recovery in the intended direction and they act on different attributes—the nonlinear penalty most on sex, rank-8 compression most on race—taking additional $A/race$ recovery from 0.0274 to 0.0055 against J ’s 0.0071 while holding residence gain at 0.0259

Quantity	Defective objective (Study 2)	Repaired objective (Study 3)
Rotation-only share of training gain, attained (<i>a property of the objective</i>)	mean 0.628, range 0.367–0.907 over 18 cells	max. abs. 1.4×10^{-15} over 18 cells
Direct invariance identity $ L(WQ) - L(W) $ on real data	up to 7.5×10^{-3}	5.6×10^{-17}
<i>Measured disclosure, repaired minus defective, over 6 matched contrasts $\times$ 4 sensitive endpoints $\times$ 2 weightings:</i>		
Sensitive cells significantly worse (more recovery)	8 of 48	
Sensitive cells significantly better (less recovery)	0 of 48	
Largest single increase in recovery	+0.0107 A / race, riv16-L2	
Residence cells resolved in either direction	0 of 12	

Table 2: The repair worked on the objective and did not work on the release. The upper rows are a diagnostic of the *objective*; the lower block is a measurement of the *release*. They are reported separately because an exactly invariant objective did not produce less measured disclosure. The attained rotation share is not a supremum – every rotation search stopped on budget exhaustion or a failed line search – and it is not a causal decomposition of what the optimiser did.

against J ’s 0.0215. But **all eight registered coordination cells failed**, the best candidate **n1r8-C1** remained significantly worse than J on local sex (+0.01065 unweighted, +0.01210 person-weighted; adjusted intervals exclude zero), an exploratory cross-year look widened rather than narrowed that gap, and the nonlinear penalty was not harmless: at rank 8 with the $L1$ policy it made A/race recovery significantly *worse*, +0.00862 [+0.00168, +0.01556] unweighted. Denominators are in Appendix D; most of the grid is simply undecided at three seeds.

Study 2 then diagnosed, after those maps had been evaluated, that its penalty was *coordinate-dependent*: summing degree-two monomials $a \leq b$ with equal weight computes $\sum_a M_{aa}^2 + \sum_{a < b} M_{ab}^2$ rather than $\|M\|_F^2$, and only the latter survives $M \mapsto Q^\top M Q$. A separately optimised, information-preserving rotation of the original solution’s subspace reaches a mean 0.628 of the refinement’s training-objective gain (range 0.367–0.907 over 18 conditions). That share is **attained, not a supremum**—every rotation search stopped on budget exhaustion or a failed line search—and it is a property of the objective, not an accounting of what the optimiser did.

Study 3 repaired it exactly, and the repair made the measured outcome worse. The repaired objective is a function on the Grassmannian: rotation-only share 1.4×10^{-15} over 18 cells against a registered forecast of 0.01, and $|L(WQ) - L(W)| \leq 5.6 \times 10^{-17}$ on real data. This gate was computed after fitting and *before* any 2018 or 2017 score was read, which is why the repair claim is supportable at all. Under the repaired objective—the only common yardstick—the repaired map attains a lower value than the defective one in 18 of 18 matched cells. And its releases disclose *more*: over six matched contrasts $\times$ four sensitive endpoints $\times$ two weightings, **8 of 48 cells are significantly worse and none is significantly better** (Table 2). The largest is +0.0107 [+0.0034, +0.0180] on local race at rank 16; none of the twelve residence cells resolves in either direction, so the extra disclosure buys no measured capability. Of Study 3’s seven registered forecasts, three were wrong, and the authors’ own registration of the central one was ill-posed; Appendix E reproduces all seven.

What this does and does not show. It refutes the prediction made from the rotation diagnostic—that removing the provably inert slack would let the same compute buy more protection. It does *not* show that rotation invariance is harmful, and it does not isolate a cause. The repair changed five things at once: Frobenius weighting, a single scalar block scale in place of per-feature standardisation, no centering, an exact radial kernel in place of a finite Fourier bank, and frozen per-role 512-row subsets. No ablation separates them, and the kernel substitution changes the estimand—the new block averages over at most 512 rows and carries a measured $1/m$ V-statistic floor. Surrogate-attacker mismatch remains the live explanation; “coordinate dependence was an implicit regulariser” is a hypothesis this design cannot test, and we do not assert it. Nor does four failed coordination cells in one construction show that nonlinear conditional-dependence penalties cannot satisfy a coordination rule. Finally, the repaired arms are significantly *worse* than their own linear-moment controls on A/race at rank 8 in three contrasts, so “the repaired arms beat their own controls”—true of exploratory 2017 seed means—is not true of the 2018 intervals.

Arm	wt.	additional recovery vs. riv16-C1				residence	source
		A / sex	AB / sex	A / race	AB / race	loss diff.	allow.
LEACE	unw.	−0.0081▼	−0.0086▼	−0.0131▼	−0.0158▼	+0.0036	3/3
	PWGTP	−0.0097▼	−0.0093▼	−0.0154▼	−0.0166▼	+0.0040	
SPLINCE	unw.	−0.0088▼	−0.0077▼	−0.0183▼	−0.0165▼	+0.0089▲	3/3
	PWGTP	−0.0100▼	−0.0084	−0.0201▼	−0.0163▼	+0.0091▲	
OptNet-C1	unw.	−0.0074▼	−0.0110▼	−0.0161▼	−0.0145▼	+0.0086▲	2/3
	PWGTP	−0.0089▼	−0.0117▼	−0.0183▼	−0.0151▼	+0.0076▲	
OptNet-L1	unw.	−0.0032	−0.0048	−0.0056	−0.0060	+0.0057	2/3
	PWGTP	−0.0059	−0.0069	−0.0116	−0.0075	+0.0046	
OptNet-L2	unw.	−0.0104▼	−0.0092▼	−0.0184▼	−0.0198▼	+0.0096▲	2/3
	PWGTP	−0.0125▼	−0.0112▼	−0.0210▼	−0.0212▼	+0.0082▲	

Table 3: Adapted published methods against the repaired mechanism at rank 16 with the coalition policy. Every arm transforms or replaces *only* the auxiliary channel and appends it to unchanged H_A . ▼ adjusted simultaneous interval entirely below zero (the arm recovers *less* than riv16-C1); ▲ entirely above. Residence is a *log loss* difference, so a positive value means the arm supplies *less* residence capability. ACS 2018 development pools; three seeds. The last column counts seeds clearing the legacy service-probe allowance, out of 3 (riv16-C1 clears 1/3).

8 The first executed external comparisons

Studies 1 and 2 ran no published method. Study 3 did, and the answer is unambiguous in one direction and carefully bounded in the other.

What each adaptation is, and what it is not. LEACE [Belrose et al., 2023] and SPLINCE [Holstege et al., 2025] are applied to the 16-coordinate auxiliary channel *alone* and the transformed channel is appended to unchanged H_A . They are deliberately *not* applied to the concatenated $[H_A, Z]$ wire: LEACE’s P^* is a single oblique projection over all coordinates and nothing constrains it to act as the identity on H_A , so editing the joint wire would generally change the published service output. The consequence for interpretation is the important part. **Their empirical linear guarantee applies to the fitted erasure of that channel, on the moments it was fitted on. It is not a guarantee about arbitrary nonlinear prediction from the concatenated service-plus-channel view an attacker actually holds**—LEACE’s own authors conjecture that nondestructive editing against a general nonlinear adversary is intractable, and a fixture in this study confirms the boundary concretely: after erasure a linear probe reads $R^2 < 10^{-8}$ while a quadratic feature of the same erased channel still recovers. SPLINCE additionally preserves covariance with the *authorised training* tasks, not with the held-out residence task; that is an adaptation, and an asymmetry in SPLINCE’s favour on utility, since no other arm sees a task label during representation fitting. SPLINCE also shares LEACE’s kernel, so under its own Theorem 2 the two coincide after refitting an unregularised unique-minimiser model—any difference measured here comes from attacker regularisation and nonlinearity, never from a stronger guarantee. OptNet-ARL’s multi- λ multi-attribute form is an **extrapolation**: the paper asserts multi-attribute generalisation in one sentence with no equation, no normalisation rule and no code. And the project’s marginal spectral arms turned out to *be* SARL-style rebuilds—reconstructing them from the derivation reproduces the stored maps bitwise—with one measured mismatch that is not absorbable into $\bar{\lambda}$: per-attribute trace reweighting, moving the selected subspace by projector Frobenius distance 0.10–0.20. They are credited and reused, and six duplicate fits were avoided. **Shared policy coefficients do not imply equal effective regularisation across methods.**

The result. Adapted LEACE, OptNet-C1 and OptNet-L2 recover significantly less than the repaired spectral mechanism on all four sensitive endpoints under *both* weightings, and adapted SPLINCE on seven of those eight cells (Table 3; its coalition-sex cell is unresolved under person weighting). OptNet-L1 separates on none. **A closed-form linear eraser from 2023, applied to a channel this project already had, recovers less than the mechanism this line of work developed across three studies.** That is the consequential finding, and it is a finding about our mechanism.

The utility side must be stated with equal care, and the incoming study’s own summary overstated it.

Arm – J	wt.	A / sex	AB / sex	A / race	AB / race	residence
LEACE	unw.	+0.0055	−0.0020	+0.0019	−0.0022	−0.0005
	PWGTP	+0.0056	−0.0016	+0.0034	−0.0027	−0.0010
SPLINCE	unw.	+0.0047	−0.0011	−0.0034	−0.0028	+0.0048
	PWGTP	+0.0052	−0.0007	−0.0013	−0.0025	+0.0041
OptNet-C1	unw.	+0.0061	−0.0044	−0.0012	−0.0009	+0.0045
	PWGTP	+0.0064	−0.0041	+0.0005	−0.0012	+0.0026
OptNet-L1	unw.	+0.0104▲	+0.0017	+0.0093	+0.0077	+0.0017
	PWGTP	+0.0094▲	+0.0008	+0.0073	+0.0064	−0.0004
OptNet-L2	unw.	+0.0031	−0.0026	−0.0035	−0.0062	+0.0056
	PWGTP	+0.0028	−0.0035	−0.0022	−0.0073	+0.0032
riv16-C1	unw.	+0.0135▲	+0.0066	+0.0149▲	+0.0136▲	−0.0041
	PWGTP	+0.0152▲	+0.0077	+0.0188▲	+0.0139▲	−0.0050
riv8-C1	unw.	+0.0121▲	+0.0058	+0.0063	+0.0049	−0.0049
	PWGTP	+0.0147▲	+0.0070▲	+0.0099	+0.0078	−0.0052

Table 4: The same arms against the frozen neural channel J . A blank cell is an **unresolved** difference at three seeds, not a demonstrated equality: the adjusted interval contains zero and also contains effects of the size this table reports elsewhere. No equivalence or non-inferiority test was run for these cells. Negative residence means the arm supplies *more* residence capability than J .

LEACE’s residence-loss difference against the repaired arm is +0.0036 [−0.0028, +0.0100] unweighted and +0.0040 [−0.0031, +0.0110] person-weighted. That is *unresolved*, not costless: the interval admits a cost of up to 0.010 nats, larger than the recovery advantage LEACE wins on three of the four endpoints, and it contains SPLINCE’s +0.0089, which the same analysis calls significant. Against the channel LEACE actually transformed, the cost is real: +0.0098 [+0.0036, +0.0159] unweighted and +0.0078 [+0.0011, +0.0145] person-weighted, with mean residence gain over H falling from 0.0318 to 0.0220. **Linear erasure of this channel buys its protection with measurable authorised capability.** We therefore do not say LEACE dominates anything.

Against J , LEACE, SPLINCE, OptNet-C1 and OptNet-L2 are *not separated* at three seeds on any of the five family endpoints, while the repaired spectral arm is significantly worse on three of four (Table 4). “Not separated” is not equivalence: no equivalence or non-inferiority test was run for these cells, and all three external point estimates on local sex lean towards J , two of them missing significance by less than 0.0007. OptNet-L1 is significantly worse than J on local sex, so the OptNet family is not uniformly indistinguishable from it either.

One structural finding worth keeping. Linear erasure deletes exactly $\text{rank}(\Sigma_{XZ})$ directions, so the realised width of the released channel is 6, 6, 7 across the three seeds. Seed 2 differs because one race category has *no support in that seed’s own representation-fitting pool* (counts 1, 1, 0); the category is not absent from California, and a class without support is unmeasured, not protected. **Protected-class support in the fitting pool determines the realised width of a released channel** (Appendix E).

9 Study 4: direct adversarial refinement, and what its one positive does and does not support

What was run. Study 4 (evidence commit 69e790af) asked whether training the auxiliary channel directly against a refreshed ensemble of differentiable attackers—with residual-logit attackers built on a frozen service-only baseline, an explicit incremental-gain penalty, and local or coalition conditioning—improves the utility/disclosure frontier beyond J , the closed-form erasers and OptNet-ARL. Nothing in that ingredient list is new [Madras et al., 2018, Belrose et al., 2023]; the question was empirical. **Every one of its 114 fitted trajectories was initialised from A_0 , not from J :** 75 are parameter-for-parameter the frozen A_0 mapper, and 39 are that mapper followed by a deterministic PCA to width 8. It used the same general neural channel family as J but never loaded J ’s weights, so it is not a fine-tune of J , and J -initialised fine-tuning remains untested. The objective was Z-only source cross-entropy plus teacher distortion at a coefficient fixed

One panel per sensitive endpoint; no score is averaged across them. Filled markers clear the service-probe allowance in 3/3 seeds, hollow do not.

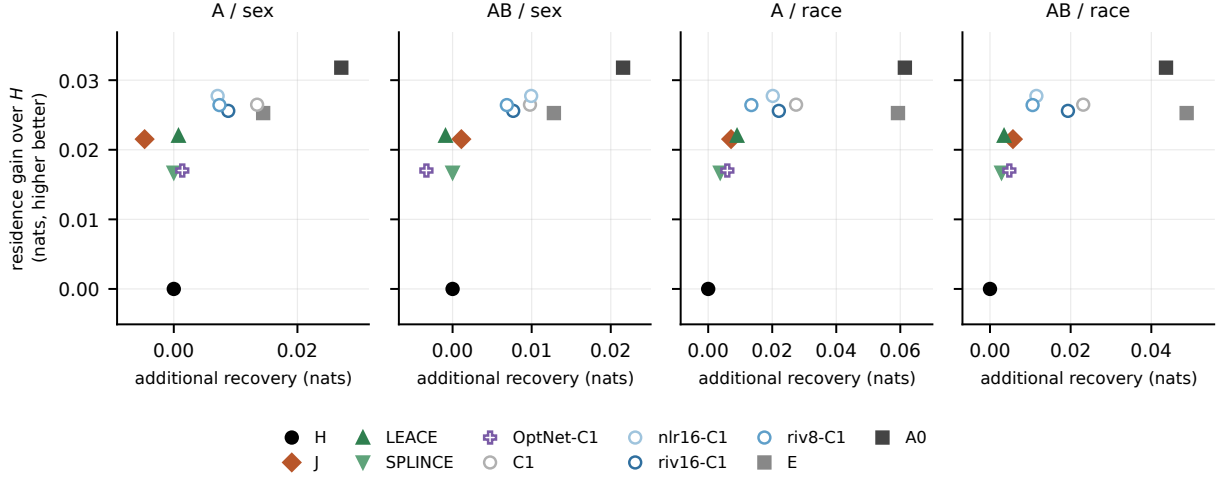


Figure 3: Authorised capability against additional recovery, one panel per sensitive endpoint. No privacy score is averaged across endpoints: doing so would let a favourable sex result hide an unfavourable race result, which is exactly what happens for several arms here.

by specification at 1.0, plus β times the policy penalty, $\beta \in \{0.1, 0.3, 1, 3\}$, two widths, three policies (L_1 ; L_2 at twice L_1 's mass; C_1 with coalition roles), three anchor seeds, repeats, single-attacker ablations and erasure controls.

The headline is negative, on both years. The registered nominee conjunction—a sensitive endpoint improved against both local controls *and* a strong external comparator—is unsupported in all 16 coordination cells. Across the 400-row frontier family, with simultaneous intervals taken over the full 850-row candidate-wide family, 189 cells are significantly worse and 18 significantly better, and all 18 are on residence utility (Table 5, Figure 4). The 2017 look reproduces the ordering: J again sits below every new arm, whose recovery on A/SEX is roughly 3–10 times J 's for about 0.006 nats more residence (Appendix G). That 2017 panel is exploratory development on a spent partition. We do not compute intervals on it; computing them would not be forbidden, but they would be conditional descriptive sampling uncertainty and would neither restore confirmatory status nor account for adaptive selection.

The positive, at the level it actually holds. The study reports that coalition conditioning beats strengthened local controls. Table 6 re-states that family at *both* correction levels and adds the one strength-matched contrast the study's own summary omits. Against the *weak* control L_1 the benefit is robust: all four sensitive endpoints at width 8 under both corrections and both weightings, and three of four at width 16 under the candidate-wide correction. Against the *strength-matched* control L_2 it is one endpoint ($AB/\text{RAC1P}$) at width 16 at the family-adjusted level, none under the candidate-wide correction the frontier results are held to, and none at all at width 8. The accompanying residence difference against L_2 is +0.00346 with family-adjusted interval $[+0.00008, +0.00684]$: the interval *crosses* the 0.001 reference, so the study's statement that it lies entirely above that reference is incorrect, and noninferiority at 0.001 is established in neither direction. Under the candidate-wide correction that interval contains zero. Lower recovery bought with lower retained capability is a trade-off, not dominance.

Why it failed—what the records do and do not show. Four observations, none causal. (a) *Selection, not stasis.* 63 of 114 trajectories (39 of 72 main arms) published their step-0 checkpoint, a channel bitwise identical to A_0 ; every one of the 63 had moved (median relative parameter displacement 0.099), and step 0

condition	role	A/SEX	AB/SEX	A/RAC	AB/RAC	residence
A0	no protection	+0.0317	+0.0294	+0.0525	+0.0469	+0.0318
J	frozen neural channel, strongest developed	+0.0015	+0.0080	+0.0000	−0.0006	+0.0215
leace_A0	closed-form eraser on A_0	+0.0039	+0.0038	+0.0097	+0.0096	+0.0220
splince_A0	closed-form eraser on A_0	−0.0002	+0.0024	+0.0081	+0.0073	+0.0167
optnet16_C1	published-method adaptation	+0.0070	+0.0046	+0.0060	+0.0116	+0.0170
spectral_C1	Study 3, previous line	+0.0191	+0.0177	+0.0294	+0.0285	+0.0265
dax8_C1_b300	Study 4, best new arm	+0.0072	+0.0096	+0.0047	+0.0079	+0.0220
dax16_C1_b100	Study 4, coalition width 16	+0.0114	+0.0162	+0.0201	+0.0182	+0.0226
dax16_L2_b300	Study 4, strengthened local	+0.0104	+0.0173	+0.0118	+0.0165	+0.0173
leace_dax8_none	erasure at rank 0	+0.0000	+0.0032	+0.0001	−0.0011	−0.0005

Table 5: Study 4 on the 2018 development pools. The best new arm, **dax8_C1_b300**, is significantly worse than **J** on **AB/RAC1P** under both weightings and significantly worse than **leace_A0** on **AB/SEX**, and significantly better than no comparator on any sensitive endpoint. Seed means over three anchor seeds, matched-exposure scope **kernel_expanded_independent**, audit budget 360, unweighted. Recovery is additional recovery over H ; residence is utility gain over H . Lower recovery is less disclosure, higher residence is more capability. Recomputed cell by cell from **PER_SEED.csv**; all 50 cells agree with the study’s own table to within 5×10^{-5} .

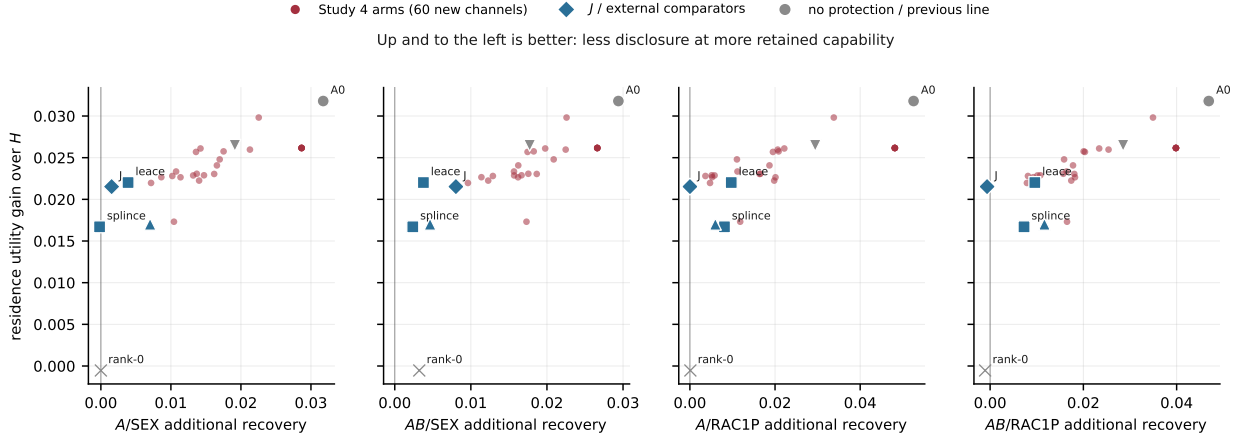


Figure 4: Residence capability against each sensitive endpoint separately, 2018 development, unweighted seed means. Red points are the Study 4 arms; bitwise-duplicate arms overplot. Comparing point estimates, no Study 4 arm lies up and to the left of **J** on *any* panel. Under person weighting (not drawn), one of the 38 arm labels does on the **AB/SEX** panel and none does on all four. These are point comparisons; significance is in Table 5’s caption and the text.

usually won by a margin well above a near tie. Optimisation moved the channel and the registered selection objective preferred the starting point. Teacher distortion is minimised at its own reference, but the full objective is not: the protection penalty at step 0 is the *largest* on the trajectory, and across the 51 arms that did move, movement cost +0.014 in source loss and +0.028 in distortion to buy −0.084 in $\beta \times$ penalty. Unchanged selected channels therefore do not diagnose a cause, and the distortion coefficient was specified, never shown optimal. Rescoring the stored trajectories at $\gamma = 0$ changes 50 of 114 selections; that is counterfactual *selection* on fixed trajectories, not retraining at $\gamma = 0$ (Figure 5). We reproduced every one of these numbers from the stored fit records. (b) *The attacker machinery was not shown to help in this run.* Refreshed attackers were kept in 84 of 1350 decisions and extended catch-up gained 2×10^{-5} ; a 720-epoch stress continuation won selection in 0 of 72 role-cells. That bounds what the declared finite family found inside its budget; it does not show the incumbents found everything recoverable, and it is not an impossibility result for stronger ensembles. (c) *Compression explains part of the improvement:* width 8 beats width 16 on race recovery and gives up residence, an effect available with no adversarial objective. (d) *The training probe and the independent audit agree moderately* (Pearson 0.58 over 900 role-cells), so the mapper was not merely fooling its own attacker family.

contrast		sensitive endpoints better, of 4		residence difference		
C_1 arm	local control	family-adj.	cand.-wide	estimate	family-adj. interval	cand.-wide
dax16_C1_b100	dax16_L1_b100	4	3	+0.00717	[+0.00305, +0.01129]	worse
dax16_C1_b100	dax16_L2_b100	1	0	+0.00346	[+0.00008, +0.00684]	n.s.
dax8_C1_b100	dax8_L1_b100	4	4	+0.00310	[+0.00041, +0.00579]	n.s.
dax8_C1_b100	dax8_L2_b100	0	0	+0.00002	[-0.00233, +0.00237]	n.s.

Table 6: Coalition against local conditioning at $\beta = 1$, at both correction levels. Unweighted, $\beta = 1.0$, matched-exposure scope. *Family-adjusted* is the single-step max- $|t|$ interval inside the `coalition_vs_local` family; *candidate-wide* is the simultaneous studentized Bonferroni bound over the full 850-row family, which is the standard the frontier comparison in Table 5 is held to. The study’s own report gives this family at the family-adjusted level only and omits the fourth row, which is the width-8 comparison against the strength-matched control and is null throughout. The coalition benefit is robust against the weak control L_1 and does not survive against L_2 under the candidate-wide standard. L_2 matches C_1 ’s nominal total group weight, not its gradients and not its difficulty.

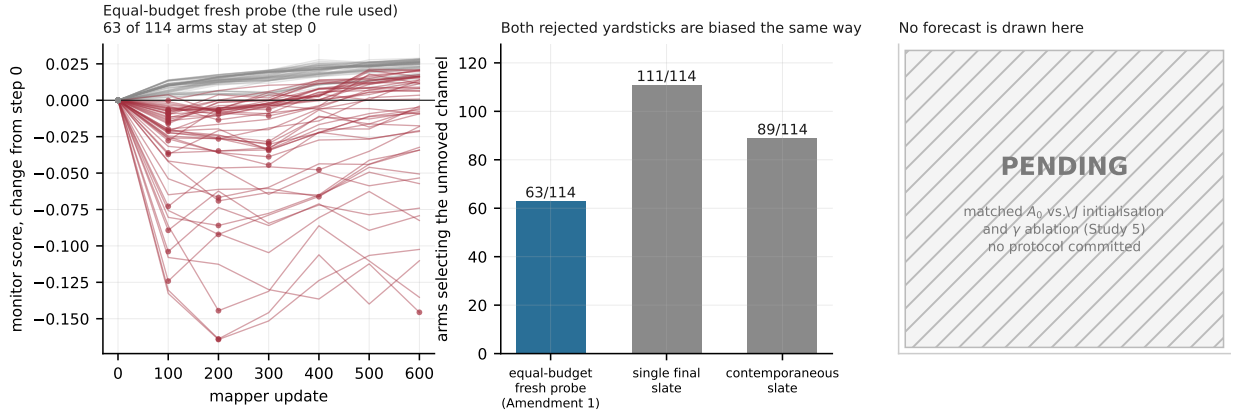


Figure 5: Left: monitor score along each of the 114 trajectories relative to step 0 under the equal-budget fresh-probe rule the study used; grey trajectories published step 0. Centre: how many trajectories each selection yardstick would return unmoved. The single final slate would have returned 111 of 114 and the contemporaneous slate 89; both are biased towards never moving, which is why the study replaced them before opening any outcome. Right: the matched initialisation and γ ablation is Study 5 and has no data.

Two results that outlast the headline. First, the fifteen external-baseline transports Study 3 could not run now exist: `leace_A0`, `splince_A0` and three OptNet policies were reconstructed and each was admitted only after rebuilding its 2018 release bitwise, at maximum absolute difference exactly 0 in all 15. On 2017, `splince_A0` is the strongest protection arm on AB/SEX and $A/RAC1P$ at the lowest residence gain. The finding that a closed-form eraser applied to a channel this project already had matches J has now been seen on two years; both looks are development. Second, the study found that its inherited attack scope gave historical arms catch-up attacks no new arm received: an arm bitwise identical to A_0 scored 0.0155 nats *better* than A_0 on $A/RAC1P$ under that scope and exactly equal under the matched-exposure scope. We report the change of scope as an evaluation correction that removes an artefact of unmatched attack opportunity, not as a gain in protection, and the exact-equality row as the study’s strongest end-to-end correctness check.

Accounting. 126 nominal fits produced 123 releases (three width-8 SPLINCE arms are scoped infeasible), 66 distinct channels and **60 distinct, new channels** (Appendix Table 11); duplicates are outcomes of the selection rule, reused with proof. LEACE at width 8 released a constant channel: measured retained rank 0 with post-erasure cross-covariance near 10^{-35} in every seed. That is a measurement. The reason the study gives—more target coordinates than feature coordinates—does not by itself imply it: at width 16 the measured rank loss is 10, 10 and 9 across seeds against a nominal 10, and one RAC1P class has a single complete case in a fit fold (Table 12).

track	arm family	nominal slots	result	precondition checked before filling
neural factorial	initialisation A_0 vs. J , common frozen A_0 teacher	168	PENDING	actual J weights loaded and verified; teacher held common
	teacher coefficient γ , penalty β		PENDING	$\gamma = 0$ removes preservation, retains authorised source losses
	local vs. coalition policy		PENDING	matched contrasts; comparison with untouched J , not only with A_0
projection	coalition-conditioned rank reduction on frozen A_0/J	204	PENDING	measured cross-moment rank and per-class support reported
	local-conditioned, local-expanded, marginal partial		PENDING	mass-scaled local hard projection registered as an alias, not a comparator
	PCA, random compression, ordinary erasers		PENDING	compression is a live explanation of Study 4's width effect

Table 7: Study 5 cells. **No result from either track has been read, and no protocol for either has been committed.** At the time of writing the branch carrying this program is byte-identical to the Study 4 evidence branch. The slot counts are the nominal registry; they are not distinct models and they are not an execution record. Study 4's own registry resolved 126 nominal slots into 123 releases, 66 distinct channels and 60 distinct-and-new, and the same three-way accounting is required here. The finite protocol permits *prospective* reductions on measured runtime; retrospective ones would be a different thing. Cells are empty because they are empty.

10 The pending programme, reviewed before its outcomes

Terminal-side experimental work continues on two tracks: (N) a factorial neural experiment varying initialisation A_0 versus J against a *common* frozen A_0 teacher, the teacher coefficient γ , β , and local versus coalition policy; and (P) a coalition-conditioned rank-reducing projection applied to frozen A_0 or J , against local-conditioned, local-expanded and marginal partial projections, PCA, random compression and ordinary erasers. **At the time of writing neither track has a committed protocol, and no outcome exists** (Table 7). What exists is a selection diagnosis of Study 4 that we verified, and our own mathematical review of the construction as specified.

Mathematical review, in brief. Appendix H gives the detail and `MATH_FIXTURES.json` the executable fixtures. The construction is well defined as an *oblique* rank- k projection, provided its range lies in the covariance support and the moment is taken on the declared masks with a globally centred channel—per-role re-centring makes the moment zero by construction. The coalition basis can detect information the local one cannot, because H_{AB} enlarges the conditioning set; duplicating labels does not. A mass-doubled local hard projection is an exact alias of the local one and is not a comparator. The convex-correction claim is true only for a fixed offset, a fixed linear basis with no intercept, and the masked rows scored; its smallest counterexample is an intercept against an offset fitted on a disjoint fold. Zero finite moments coexist with perfectly recoverable S . With full-rank Σ the map is exactly LEACE's closed form with a different cross-moment; with rank-deficient Σ it deletes off-support directions that LEACE would keep, so the form must be declared. Algebraically the construction is LEACE's structure applied to a coalition-conditioned cross-moment on strong existing channels with explicit rank controls—an adaptation, which we will call one unless a full prior-work comparison supports more. No arbitrary-nonlinear privacy theorem is needed to measure whether it helps.

What each level of claim would require. We keep four levels apart. *Mechanism correctness*: the fixtures above, plus measured rank, support and tolerance for every arm. *Competitive development trade-off*: a candidate chosen without residence or commute labels or evaluation scores, one configuration shared by all three anchor seeds, compared with untouched J , executed LEACE/SPLINCE/OptNet adaptations and the local controls, reporting superiority, noninferiority at the declared 0.001 utility margin and 0.001-nat sensitive tolerance, and the zero-margin result, each by name—a tolerance is not zero harm, and nonsignificance is not equivalence. *Coalition-specific benefit*: the same source channel under local conditioning, at matched rank and capacity, at one declared correction level. *Independent confirmation*: a fresh prospective protocol on a partition no study has used; 2016 is sealed, unscored and not opened by any development result. Post-hoc frontier observations may be reported if labelled and corrected for the comparisons searched; they are not the prospective candidate.

11 What this evidence does not establish

Stated once, in one place, rather than scattered.

No attack maximum is an upper bound. Every recovery number is what a *validation-selected* member of a finite attack collection achieved. It is a floor. Our own data contain a clean demonstration: when J is given additional attack candidates, its measured absolute recovery *falls*, because selection on validation loss can pick a candidate that generalises worse (Appendix F). A stronger attacker we did not build would move any of these numbers upward.

Nested validation-selected models need not produce non-negative gains. Statement 2 of §3 is about population optima. In finite samples, 22 of 429 unweighted role-cells have *negative* increments, and J 's own local-sex increment is negative under person weighting with an interval excluding zero. We report these unclipped, because clipping them would manufacture protection out of a selection artifact. H 's own attack, routed onto an augmented wire, remains executable, so a negative increment removes nothing an H -only attacker already had.

No detected difference is not equality. Most of the grid in every attribution family is unresolved at three seeds. We mark such cells blank and never read them as parity.

Ratios are scale-dependent; paired differences are not. $C1$ against J on A/race is $6.0\times$ on increments and $2.6\times$ in absolute terms; the paired difference is identical on both scales because the shared H baseline cancels inside a paired contrast. We therefore quote differences. This also makes every headline contrast in this paper invariant to the choice of attack scope, even though the H -relative *levels* are not (Appendix F).

Nothing here is a certificate. Vanishing fitted finite moments imply neither $Z \perp S \mid H$ nor any bound on $I(S; Z \mid H)$; no Type-I control is inherited from KCI or RCoT; and a misspecified nuisance manufactures a penalty more than ten times the oracle value under exact conditional independence.

Development against confirmation. Only the 2017 transport is confirmatory. The 2018 pools have been used repeatedly; the 2017 seal is spent, so the cross-year passes of Studies 3 and 4 are development. Study 4 completed the erasure and OptNet transports Study 3 had excluded. Neither pass carries intervals; that is a reporting choice, not a prohibition—conditional sampling intervals on a used partition are descriptive, and do not restore confirmatory status or account for adaptive selection. **ACS 2016 remains unscored:** admitted on provenance and schema with a prospective label-blind split prepared, with no label, output, transform or performance figure inspected. A positive development result may motivate a confirmation; it can never inherit a sealed year's status.

Sampling and design. Three seeds are three fixed fitted systems on one cohort. The bootstraps cover people, not retraining variability and not Census design variance. Transport is between public survey years of the same state and age band, and cross-year person distinctness cannot be established from public identifiers. Income is nominal and unadjusted and its estimand differs again in 2016.

Multiplicity is only as honest as its level. A positive reported at a family-adjusted level beside negatives reported at a candidate-wide level overstates the positive. Study 4 did this; we re-state its coalition family at both levels (Table 6), and we ask the pending programme to declare one level before fitting.

A tolerance is not zero harm. The pending programme's 0.001-nat sensitive tolerance and 0.001 utility margin are noninferiority margins. A result inside them is noninferior at that margin, not harmless, and we will report the zero-margin result beside it.

Numerical integrity. Study 3's exploratory 2017 stage aborted four times with a probability row summing to exactly zero inside an otherwise well-formed array. Every occurrence was detected by full-schema validation and aborted its unit before any metrics file was written; the four affected units were quarantined and recomputed by the unpatched pipeline, so no corrupted array reached any table. The machine was at swap capacity throughout. **Memory pressure was the observed condition, not a demonstrated cause**, and nothing here claims the machine is now proved reliable. One document in the incoming package records that no such fault occurred; it predates the stage in which they did, and we follow the run ledger instead (CORRECTIONS.md B10). Study 4's 2017 stage recorded four more such transient probability-validation failures, each caught before any metrics file was written and recomputed cleanly on retry, again at near-zero free memory; the cause is not established.

Claim	Status	Supporting measurement	What it does not establish
H preserved exactly in every release	Structural identity	Verified bitwise, 210/210 views	Says nothing about service <i>accuracy</i> , which moved by up to 0.015 nats
Appended channel adds authorised capability	Empirical, development & sealed	0.017–0.032 nats residence gain over H	Half-headroom reference fails in 1–3 of 3 seeds for every arm
Coalition penalty beats both local controls	Confirmatory (2017 sealed)	14/16 cells better, 0 worse, 2 unresolved	One year, one state, three seeds; residence cost not bounded below 0.001
Developed mechanism vs. frozen neural channel	Empirical, development	Worse on 3 of 4 sensitive endpoints	J is internal, and it supplies <i>less</i> residence capability: neither dominates
Rotation defect repaired exactly	Elementary algebra + numerics	Invariance to 10^{-15} , 18/18 cells	Invariance of an objective is not evidence the objective measures the right thing
Repair reduces measured disclosure	Refuted	4 cells significantly worse, 0 better	Attributable to the repaired <i>package</i> , not to any single changed ingredient
External adaptations beat the developed mechanism	Empirical, development	LEACE / SPLINCE / OptNet-C1 lower on all four endpoints	Adaptations, not replicas; LEACE costs residence against its own source channel
Any bound on $I(S; Z H)$	Not claimed	–	Finite validation-selected attacks attain no infimum in either direction

Table 8: Every load-bearing claim, its status, the measurement behind it, and what it does not establish. The full traceable list, with source commit, artifact, endpoint, data pool and uncertainty procedure for each, is `CLAIM_LEDGER.csv`.

12 Claims this revision withdraws or narrows

Thirty-one items corrected in earlier revisions are carried forward unchanged. The v4 additions, all additive to the historical records (`CORRECTIONS_V4.md`), are these; earlier ones follow.

1. **Study 4 as “a refinement of J ”.** All 114 trajectories were initialised from A_0 . J -initialised fine-tuning is untested.
2. **“Both residence intervals lie entirely above zero and above the 0.001 reference.”** False for the L_2 contrast: $[+0.00008, +0.00684]$ crosses 0.001, and its candidate-wide interval contains zero.
3. **“Coalition conditioning genuinely beats strong local controls.”** Narrowed: robust against L_1 ; against the strength-matched L_2 , one endpoint at one width at the family-adjusted level only, and null at width 8—a row the original summary omitted.
4. **“ A_0 is very nearly the minimiser of U by construction.”** Only the distortion term is minimised at the teacher. Unchanged released channels are a selection outcome; every such trajectory had moved.
5. **“The incumbents were genuinely strong.”** Narrowed to: the declared finite family, inside its budget, found little more. Not a statement that everything recoverable was found.
6. **Rank-zero LEACE “because $11 > 8$ ”.** The nominal count is 10, and counting does not determine the projection; the result stands as a measurement.
7. **“39 of 72 unmoved at $\beta \in \{0.1, 0.3\}$ ”.** Three of the 39 are at $\beta = 1$.
8. **The scope change of Study 4’s amendment 4** is an evaluation correction, never a protection gain.

The earlier corrections, including the withdrawal of “LEACE dominates” and of equivalence to J , are listed in Appendix I.

13 Conclusion

A service that has already published predictions can add a reusable channel beside them without changing a released number, and can steer that channel away from directions a recipient or a colluding pair would use to infer sex and race. On a locked, previously unused survey year, the coalition-conditioning knob does what it claims against both of its local controls—14 of 16 sensitive cells strictly better, none worse, two unresolved—while costing residence capability by an amount the data cannot bound below the 0.001 nats the registered rule used. Everything built afterwards to make the mechanism competitive failed: a nonlinear penalty failed all eight of its coordination cells and made one endpoint significantly worse; an

exact repair of that penalty’s coordinate dependence—the one intervention the diagnostic predicted would help—produced significantly *more* disclosure on eight of forty-eight cells and less on none; the first executed external comparisons put three adapted published methods below the developed mechanism on every sensitive endpoint; and direct adversarial refinement from A_0 produced 60 new channels, none of which improves a sensitive endpoint against any comparator, with its coalition advantage surviving the simultaneous correction only against the weak local control.

The defensible contribution is the release problem and the evaluation design: fixed published outputs as a hard constraint, recipient-specific and coalition-specific audit roles, fresh and frozen attackers never pooled, absolute and incremental recovery reported side by side, negative increments retained unclipped, denominators printed beside every count, per-endpoint results never averaged into one score, and a sealed temporal transport with the multiplicity fixed in advance. The measurement contract is what made three successive negative results legible instead of ambiguous. The mechanism family, as tested so far, is not competitive, and we say so.

How this conclusion will change. Exactly one of two paragraphs replaces this one once the pending programme reports. *If a candidate earns the competitive-development label* under the criteria of §10, we will describe its exact construction, its advantage over J and the executed external adaptations endpoint by endpoint, its utility cost and the margins used, and the confirmation it still requires; the four negative studies become the evidence for its design choices, not a proof of its novelty. *If neither track earns it*, the empirical contribution stands as above and we will say which of the proposed remedies— J initialisation, a smaller teacher coefficient, coalition-conditioned rank reduction—failed and how. Neither outcome has been observed, and neither closes method development.

14 Reproducibility

Deterministic seeds throughout; every unit writes a hash-bound completion record. All 90 pre-declared Study 1 endpoints were regenerated from stored per-person predictions by an independently written scorer agreeing to 1.1×10^{-15} . Study 3’s own independent score replay reports a maximum absolute difference of 4.4×10^{-16} over 3,264 checks, and H_A/H_B parity is bitwise on all seven pools for every arm at release time and again at evaluation time. Every table and figure here is generated by two scripts from committed evidence; the Study 3 assets are read at a pinned commit through the object store, so each printed digit traces to a (commit, path, sha256) triple in MANIFEST.json. No number was typed by hand. Commands and file map: REPRODUCE_PAPER.md in results/pcrl_manuscript_review_v3/. For v4, every Study 4 number was recomputed from PER_SEED.csv, PAIRED_INTERVALS.csv and the 114 stored fit records rather than copied (results/pcrl_manuscript_review_v4/checks/); the frontier table agrees in 50 of 50 cells, the 2017 panel cell for cell, and Study 4’s own score replay reports 4.4×10^{-16} over 4,512 checks. The mathematical statements of §10 are backed by executable fixtures in MATH_FIXTURES.json. This revision fitted no ACS model.

A Elementary results and their boundaries

Proofs and the exact conditions under which each holds are in `MATHEMATICAL_REVIEW.md`. Each statement is given at four levels—population definition, empirical estimator, algorithmic approximation, numerical test—because conflating them is the failure mode this appendix exists to prevent.

B Related-work comparison

Work	fixed prior output	reusable extra features	recipient permissions	coalition-aware training	independent attack eval.
This paper	✓	✓	✓	✓	✓
Madras et al. 2018 (LAFTR)	–	✓	–	–	part.
LEACE [Belrose et al., 2023]	–	–	–	–	–
SPLINCE [Holstege et al., 2025]	–	–	–	–	–
SARL [Sadeghi et al., 2019]	–	–	–	–	part.
OptNet-ARL [Sadeghi et al., 2021]	–	–	–	–	part.
K-TOpt [Sadeghi et al., 2022]	–	–	–	–	part.
U-FaTE [Dehdashtian et al., 2024]	–	–	–	–	✓
Sankar et al. [2013]	–	UNV.	UNV.	UNV.	–
Stadler et al. 2024	–	–	–	–	n/a
Elazar & Goldberg 2018	–	–	–	–	✓

Table 9: The five features this paper’s setting turns on. ✓ the text demonstrates it; – its setting excludes it; “part.” a partial form (an adversarial objective or a transfer test, not an independent attacker slate); UNV. could not be settled from what was read and is **not** guessed from a title or abstract. Several of these methods can be *adapted* into this setting; three were, and they beat the mechanism developed here. Provenance for every row, including which sources were read in full text and which only as abstracts, is in `RELATED_WORK_COMPARISON.md`.

The column that is ours is the first: no text we read *evaluated* a mechanism under the constraint that a previously published output is immutable, with per-recipient and coalition audit roles. That is absence of evidence after a bounded search, not a priority claim, and it would be false as a statement about what these methods *can* do — three of them were adapted into this setting in §8 and beat the mechanism developed here. The attribution ledger, naming every component this work reuses and from whom, is in `RELATED_WORK_COMPARISON.md`.

C Study 1: locked transport, full tables

Contrast	Endpoint	Estimate	Seeds 0 / 1 / 2	Adjusted 95%	
<i>F1 (registered primary): coalition penalty vs local controls, $c = 2.930$ over 20 endpoints</i>					
C1-L1	residence loss	+0.00029	+0.0005 / +0.0004 / -0.0001	[-0.00118, +0.00176]	
C1-L1	A / sex	-0.00282	-0.0046 / -0.0055 / +0.0017	[-0.00585, +0.00020]	
C1-L1	AB / sex	-0.00593	-0.0046 / -0.0094 / -0.0038	[-0.00915, -0.00271]	▼
C1-L1	A / race	-0.01369	-0.0145 / -0.0129 / -0.0137	[-0.01737, -0.01002]	▼
C1-L1	AB / race	-0.00792	-0.0054 / -0.0109 / -0.0075	[-0.01105, -0.00478]	▼
C1-L2	residence loss	+0.00059	+0.0012 / +0.0009 / -0.0004	[-0.00073, +0.00191]	
C1-L2	A / sex	-0.00258	-0.0020 / -0.0051 / -0.0007	[-0.00568, +0.00052]	
C1-L2	AB / sex	-0.00472	-0.0058 / -0.0060 / -0.0024	[-0.00804, -0.00140]	▼
C1-L2	A / race	-0.00741	-0.0054 / -0.0074 / -0.0095	[-0.01088, -0.00394]	▼
C1-L2	AB / race	-0.00302	+0.0004 / -0.0039 / -0.0056	[-0.00524, -0.00081]	▼
<i>F3 (registered secondary): coalition channel vs frozen neural channel J, $c = 2.728$ over 10 endpoints</i>					
C1-J	residence loss	-0.00932	-0.0141 / -0.0020 / -0.0118	[-0.01500, -0.00363]	▼
C1-J	A / sex	+0.01301	+0.0201 / +0.0034 / +0.0155	[+0.00753, +0.01848]	▲
C1-J	AB / sex	+0.00697	+0.0088 / +0.0014 / +0.0107	[+0.00151, +0.01242]	▲
C1-J	A / race	+0.03767	+0.0428 / +0.0383 / +0.0320	[+0.02941, +0.04594]	▲
C1-J	AB / race	+0.03268	+0.0330 / +0.0318 / +0.0332	[+0.02504, +0.04032]	▲

Family	Sens. cells	Better	Worse	Unresolved	advantage fired	c
F1 primary (C1 vs L1, L2; Mode B)	16	14	0	2	4 / 4	2.930
F2 replication (J vs L025, L20; Mode B)	16	6	0	10	4 / 4	3.024
F3 secondary (C1 vs J; Mode B)	8	0	8	0	0 / 2	2.728
F4 frozen transfer (all four; Mode A)	32	21	2	9	6 / 8	3.206

Contrast	Weight	Point est.	Pointwise 95% up.	NI?	Simult. 95% up.	NI?
C1 - L1	unw.	+0.00029	+0.00112	no	+0.00166	no
C1 - L1	PWGTP	+0.00010	+0.00103	no	+0.00164	no
C1 - L2	unw.	+0.00059	+0.00135	no	+0.00182	no
C1 - L2	PWGTP	+0.00017	+0.00102	no	+0.00162	no

Interface	source-probe allowance		half-headroom residence	
	Mode A (frozen)	Mode B (fresh)	Mode A	Mode B
H	3 / 3	3 / 3	0 / 3	0 / 3
E	3 / 3	3 / 3	0 / 3	0 / 3
A0	3 / 3	3 / 3	0 / 3	1 / 3
J	3 / 3	3 / 3	0 / 3	0 / 3
S0	0 / 3	3 / 3	2 / 3	1 / 3
L1	0 / 3	3 / 3	0 / 3	1 / 3
L2	0 / 3	3 / 3	0 / 3	1 / 3
C1	0 / 3	3 / 3	0 / 3	1 / 3

Comparator	gain at $p=1$	p^*	A/race at p^*	reachable?
E	0.0263	1.072	—	no
A0	0.0289	0.976	0.0638	yes
L025	0.0175	1.615	—	no
L20	0.0181	1.555	—	no
J	0.0189	1.493	—	no
S0	0.0304	0.929	0.0688	yes
C1 (target)	0.0282	—	0.0453	—

Interface	Residence gain	A / race		AB / sex	
		absolute	additional	absolute	additional
H	0.0000	0.0165	+0.0000	0.0141	+0.0000
J	0.0189	0.0241	+0.0076	0.0159	+0.0017
E	0.0263	0.0794	+0.0629	0.0299	+0.0158
A0	0.0289	0.0818	+0.0653	0.0393	+0.0251
S0	0.0304	0.0905	+0.0740	0.0397	+0.0256
L1	0.0285	0.0754	+0.0590	0.0288	+0.0146
L2	0.0288	0.0692	+0.0527	0.0275	+0.0134
C1	0.0282	0.0617	+0.0453	0.0228	+0.0087

D Study 2: attribution and denominators

Factor	Held at	Sens. cells	Better	Worse	Unres.	Residence better/worse
Nonlinear penalty	$r = 16$	24	7	0	17	0 / 0 (of 6)
Nonlinear penalty	$r = 8$	24	5	1	18	0 / 0 (of 6)
Rank-8 compression	both penalties	48	15	0	33	0 / 0 (of 12)
Coalition vs local penalty	both ranks	64	29	2	33	0 / 0 (of 16)

Endpoint	Weight	Estimate	Adjusted 95%
residence loss	unw.	-0.00435	[-0.01238, +0.00368]
A / sex	unw.	+0.01065	[+0.00331, +0.01799] ▲
AB / sex	unw.	+0.00247	[-0.00396, +0.00890]
A / race	unw.	-0.00160	[-0.01160, +0.00840]
AB / race	unw.	-0.00170	[-0.00832, +0.00492]
residence loss	PWGTP	-0.00432	[-0.01349, +0.00485]
A / sex	PWGTP	+0.01210	[+0.00441, +0.01978] ▲
AB / sex	PWGTP	+0.00422	[-0.00273, +0.01117]
A / race	PWGTP	+0.00133	[-0.00934, +0.01200]
AB / race	PWGTP	+0.00052	[-0.00713, +0.00818]

Seed	q	U : pos / neg / ≈ 0	L1 pos	L2 pos	C1 pos	r_+
0	128	32 / 0 / 96	32	32	32	16
1	128	32 / 0 / 96	32	32	32	16
2	128	32 / 0 / 96	32	32	32	16

Family	Contrast	Endpoint	Weight	Estimate	Adjusted 95%
C1	nlr16-L2-lin16-L2	A / sex	unw.	-0.00597 ▼	[-0.01009, -0.00184]
C1	nlr16-L2-lin16-L2	A / race	unw.	-0.01706 ▼	[-0.02477, -0.00935]
C1	nlr16-L2-lin16-L2	AB / race	unw.	-0.00956 ▼	[-0.01902, -0.00009]
C1	nlr16-L2-lin16-L2	A / race	PWGTP	-0.01316 ▼	[-0.02176, -0.00456]
C1	nlr16-C1-lin16-C1	A / sex	unw.	-0.00639 ▼	[-0.01056, -0.00221]
C1	nlr16-C1-lin16-C1	A / race	unw.	-0.00728 ▼	[-0.01422, -0.00034]
C1	nlr16-C1-lin16-C1	AB / race	unw.	-0.01161 ▼	[-0.02116, -0.00206]
C6	nlr8-L1-lin8-L1	A / race	unw.	+0.00862 ▲	[+0.00168, +0.01556]
C6	nlr8-L2-lin8-L2	A / sex	unw.	-0.00850 ▼	[-0.01237, -0.00463]
C6	nlr8-L2-lin8-L2	A / sex	PWGTP	-0.00634 ▼	[-0.01054, -0.00214]
C6	nlr8-C1-lin8-C1	A / sex	unw.	-0.00653 ▼	[-0.01099, -0.00207]
C6	nlr8-C1-lin8-C1	AB / sex	unw.	-0.00548 ▼	[-0.00858, -0.00238]
C6	nlr8-C1-lin8-C1	AB / sex	PWGTP	-0.00518 ▼	[-0.00876, -0.00159]
C3	lin8-L1-lin16-L1	A / race	unw.	-0.01721 ▼	[-0.02744, -0.00698]
C3	lin8-L1-lin16-L1	AB / race	unw.	-0.01808 ▼	[-0.02775, -0.00842]
C3	lin8-L1-lin16-L1	A / race	PWGTP	-0.01336 ▼	[-0.02493, -0.00178]
C3	lin8-L1-lin16-L1	AB / race	PWGTP	-0.01232 ▼	[-0.02325, -0.00140]
C3	nlr8-L1-nlr16-L1	A / race	unw.	-0.00935 ▼	[-0.01742, -0.00129]
C3	nlr8-L1-nlr16-L1	AB / race	unw.	-0.01133 ▼	[-0.02115, -0.00151]
C3	lin8-L2-lin16-L2	A / race	unw.	-0.02384 ▼	[-0.03547, -0.01221]
C3	lin8-L2-lin16-L2	AB / race	unw.	-0.01359 ▼	[-0.02501, -0.00218]
C3	lin8-L2-lin16-L2	A / race	PWGTP	-0.01811 ▼	[-0.03112, -0.00509]
C3	nlr8-L2-nlr16-L2	A / sex	unw.	-0.00453 ▼	[-0.00903, -0.00002]
C3	lin8-C1-lin16-C1	A / race	unw.	-0.02169 ▼	[-0.03196, -0.01143]
C3	lin8-C1-lin16-C1	AB / race	unw.	-0.01490 ▼	[-0.02559, -0.00421]
C3	lin8-C1-lin16-C1	A / race	PWGTP	-0.01667 ▼	[-0.02864, -0.00470]
C3	nlr8-C1-nlr16-C1	AB / sex	unw.	-0.00636 ▼	[-0.01085, -0.00187]
C3	nlr8-C1-nlr16-C1	A / race	unw.	-0.01463 ▼	[-0.02500, -0.00425]
C4	lin16-C1-lin16-L1	AB / sex	unw.	-0.00515 ▼	[-0.00911, -0.00119]
C4	lin16-C1-lin16-L1	A / race	unw.	-0.01066 ▼	[-0.01642, -0.00489]
C4	lin16-C1-lin16-L1	AB / race	unw.	-0.01427 ▼	[-0.01980, -0.00874]
C4	lin16-C1-lin16-L1	AB / sex	PWGTP	-0.00603 ▼	[-0.01069, -0.00137]
C4	lin16-C1-lin16-L1	A / race	PWGTP	-0.01203 ▼	[-0.01849, -0.00558]
C4	lin16-C1-lin16-L1	AB / race	PWGTP	-0.01502 ▼	[-0.02149, -0.00856]
C4	lin8-C1-lin8-L1	A / race	unw.	-0.01514 ▼	[-0.02051, -0.00976]
C4	lin8-C1-lin8-L1	AB / race	unw.	-0.01108 ▼	[-0.01564, -0.00653]
C4	lin8-C1-lin8-L1	A / race	PWGTP	-0.01535 ▼	[-0.02161, -0.00908]
C4	lin8-C1-lin8-L1	AB / race	PWGTP	-0.00991 ▼	[-0.01535, -0.00446]
C4	nlr16-C1-nlr16-L1	A / sex	unw.	-0.00636 ▼	[-0.01037, -0.00235]
C4	nlr16-C1-nlr16-L1	A / race	unw.	-0.01870 ▼	[-0.02509, -0.01231]
C4	nlr16-C1-nlr16-L1	AB / race	unw.	-0.02253 ▼	[-0.03232, -0.01273]
C4	nlr16-C1-nlr16-L1	A / sex	PWGTP	-0.00728 ▼	[-0.01191, -0.00265]
C4	nlr16-C1-nlr16-L1	A / race	PWGTP	-0.01865 ▼	[-0.02570, -0.01161]
C4	nlr16-C1-nlr16-L1	AB / race	PWGTP	-0.01700 ▼	[-0.02824, -0.00576]
C4	nlr16-C1-nlr16-L2	AB / sex	unw.	+0.00399 ▲	[+0.00034, +0.00764]
C4	nlr8-C1-nlr8-L1	A / sex	unw.	-0.00544 ▼	[-0.00884, -0.00204]
C4	nlr8-C1-nlr8-L1	AB / sex	unw.	-0.00702 ▼	[-0.01050, -0.00353]
C4	nlr8-C1-nlr8-L1	A / race	unw.	-0.02397 ▼	[-0.03285, -0.01510]
C4	nlr8-C1-nlr8-L1	AB / race	unw.	-0.01871 ▼	[-0.02632, -0.01110]
C4	nlr8-C1-nlr8-L1	A / sex	PWGTP	-0.00595 ▼	[-0.00977, -0.00212]
C4	nlr8-C1-nlr8-L1	AB / sex	PWGTP	-0.00645 ▼	[-0.01046, -0.00243]
C4	nlr8-C1-nlr8-L1	A / race	PWGTP	-0.02156 ▼	[-0.03232, -0.01081]
C4	nlr8-C1-nlr8-L1	AB / race	PWGTP	-0.01774 ▼	[-0.02684, -0.00864]
C4	nlr8-C1-nlr8-L2	A / sex	unw.	+0.00229 ▲	[+0.00009, +0.00449]
C4	nlr8-C1-nlr8-L2	AB / sex	unw.	-0.00428 ▼	[-0.00726, -0.00130]
C4	nlr8-C1-nlr8-L2	A / race	unw.	-0.00539 ▼	[-0.01044, -0.00033]
C4	nlr8-C1-nlr8-L2	AB / race	unw.	-0.00868 ▼	[-0.01433, -0.00302]
C4	nlr8-C1-nlr8-L2	AB / sex	PWGTP	-0.00385 ▼	[-0.00734, -0.00037]
C4	nlr8-C1-nlr8-L2	AB / race	PWGTP	-0.01018 ▼	[-0.01717, -0.00319]

E Study 3: registered forecasts, erasure rank, conditions

Seed	directions deleted	realised width	race class 3 support	cross-cov. after	idempotency	mean pres.
0	10	6	1	4.6×10^{-16}	4.9×10^{-15}	7.7×10^{-15}
1	10	6	1	3.9×10^{-16}	6.7×10^{-15}	4.9×10^{-15}
2	10	7	0	7.2×10^{-16}	3.0×10^{-15}	6.0×10^{-15}

Linear erasure deletes exactly $\text{rank}(\Sigma_{XZ})$ directions. A centred joint one-hot over sex (2), race (9) and coverage (2) has rank $1 + 8 + 1 = 10$, leaving 6 of 16. **Seed 2 differs because one race category has no support in that seed's own fitting pool**, so the centred one-hot has rank 9 and the channel keeps 7 dimensions. Support counts refer to fitting pools, not to the population: a category missing from one pool is not absent from California.

Interface	Objective	r	Fitting labels	Inf. inputs	Role
H	none (no channel)	–	none	H_A	reference
E	adversarial, frozen	16	sex, race	X	context
A0	adversarial, frozen	16	sex, race	X	context
L025	adversarial, frozen	16	sex, race	X	F2 control
L20	adversarial, frozen	16	sex, race	X	F2 control
J	adversarial, frozen	16	sex, race	X	neural ref.
S0	U only ($\lambda = 0$)	16	none	T, H_A	context
M025 / M1	$U - \lambda P_{\text{marg}}$	16	sex, race	T, H_A	context
L1 / L2 / L025	$U - \lambda P_{\text{loc}}$	16	sex, race, cov.	T, H_A	F1/F4 control
C025 / C1	$U - \lambda(P_{\text{loc}} + P_{AB})$	16	+ coal. sex, race	T, H_A	candidate
lin16-*	$U - \lambda P$ (alias)	16	as above	T, H_A	alias, not refitted
nlr16-*	nonlinear refinement	16	as above	T, H_A	dev. only
lin8-*	$U - \lambda P$, compressed	8	as above	T, H_A	dev. only
nlr8-*	nonlinear ref., compressed	8	as above	T, H_A	dev. only

F Attack scope, and what a paired contrast is invariant to

Condition	fresh independent attacks		+ historical catch-up (reporting scope)	
	absolute	additional	absolute	additional
H	0.0014	+0.0000	0.0071	+0.0000
J	0.0027	+0.0013	0.0024	−0.0047
LEACE	0.0053	+0.0039	0.0078	+0.0008
riv16-C1	0.0159	+0.0145	0.0159	+0.0088
A0	0.0342	+0.0328	0.0342	+0.0271

A / sex, 2018, unweighted, seed means. The reporting scope gives H and the frozen historical interfaces their saved-observer catch-up attacks; the new arms have none, so their *absolute* recovery is unchanged between scopes and only the *subtracted* H baseline moves ($0.0014 \rightarrow 0.0071$). Every paired contrast between two arms is therefore identical on both scales, because the shared baseline cancels; only H -relative levels and ratios are scope-dependent. The absolute recovery of J *falls* when it is given more attack candidates – a validation-selected maximum is a floor on leakage, never a bound.

G Study 4: full tables

condition	role	A/SEX	AB/SEX	A/RAC	AB/RAC	residence
A0	no protection	+0.0325	+0.0312	+0.0632	+0.0564	+0.0289
J	frozen neural channel, strongest developed	+0.0013	+0.0069	+0.0039	+0.0015	+0.0189
leace_A0	closed-form eraser on A_0	+0.0071	+0.0072	+0.0145	+0.0101	+0.0219
splince_A0	closed-form eraser on A_0	+0.0021	+0.0009	+0.0053	+0.0049	+0.0148
optnet16_C1	published-method adaptation	+0.0093	+0.0088	+0.0211	+0.0185	+0.0139
spectral_C1	Study 3, previous line	+0.0170	+0.0148	+0.0471	+0.0436	+0.0282
dax16_C1_b100	Study 4, coalition width 16	+0.0176	+0.0167	+0.0259	+0.0231	+0.0252
leace_dax8_none	erasure at rank 0	−0.0002	−0.0002	−0.0002	+0.0003	−0.0007

Table 10: Study 4 panel on 2017, scope **common_fresh**, budget 360, unweighted seed means. 66 of 69 declared interfaces scored (23 newly transported conditions $\times$ 3 seeds; references H , A_0 , J excluded from that count); the three absent are the infeasible width-8 SPLINCE arm. **Exploratory cross-year development.** The 2017 **final_evaluation** partition is spent; no intervals are computed. Computing them would not be forbidden—they would be conditional descriptive sampling uncertainty—but they would not restore confirmatory status or account for adaptive selection. The original locked 2017 transport result keeps its historical status and is neither restated nor overwritten.

quantity	count	note
nominal transform fits planned	126	the registry, not a count of systems
released interfaces	123	3 SPLINCE width-8 arms SCOPED INFEASIBLE
bitwise duplicates within a seed	57	preregistered checkpoint rule returned the unmoved channel
distinct channels	66	123 – 57
of those, bitwise identical to a historical arm	6	max abs difference 0.0; reused with proof
distinct and new	60	the number of new systems this study contributes
main arms bitwise identical to no protection	39 of 72	18 at $\beta = 0.1$, 18 at 0.3, 3 at 1.0

Table 11: From nominal slots to distinct new systems. Nominal registry slots are not distinct models. The duplicates are outcomes of the preregistered checkpoint rule—every unit ran its full 600-update budget and selection returned the unmoved initial checkpoint—not failed or abandoned fits. Arithmetic verified in both directions and against the per-seed sums. The study’s own gloss attributes the unmoved arms to $\beta \in \{0.1, 0.3\}$; three further arms select the unmoved channel at $\beta = 1.0$.

seed	arm	width	nominal rank loss	measured retained rank	post-erasure cross-cov. max
0	leace_dax16_none	16	10	6	$4.6e - 16$
0	leace_dax8_none	8	10	0	$1.9e - 35$
1	leace_dax16_none	16	10	6	$3.9e - 16$
1	leace_dax8_none	8	10	0	$3.2e - 35$
2	leace_dax16_none	16	10	7	$7.2e - 16$
2	leace_dax8_none	8	10	0	$4.0e - 35$

Table 12: Erasure rank on the new channels, measured. The nominal rank loss is $1 + 8 + 1 = 10$ centred degrees of freedom for the joint {SEX, RAC1P, public_coverage} schema. Counting does not determine the projection: at width 16 the *measured* loss is 10, 10, 9 across seeds, so seed 2 retains rank 7 where counting predicts 6. One RAC1P class carries a single complete case in the seed-0 fit fold (support [5989, 555, 90, 1, 32, 1674, 49, 1527, 596]). Rank-zero erasure at width 8 is established by measurement—retained rank 0 with post-erasure cross-covariance at 10^{-35} in all three seeds—not by the observation that $10 > 8$.

H Mathematical review of the pending projection

The projection, checked algebraically. With row vectors, $v = (z - \mu)\Sigma^{-1/2}$ and $z_{\text{out}} = \mu + vP\Sigma^{1/2}$ compose to $z_{\text{out}} = \mu + (z - \mu)M$ with $M = \Sigma^{-1/2}P\Sigma^{1/2}$. M is idempotent of the same rank as P , but it is *oblique*, not symmetric—as is LEACE’s P^* . Two conditions must be stated. If Σ is rank deficient the pseudoinverse gives $\Sigma^{1/2}\Sigma^{-1/2} = \Pi_{\text{supp}} \neq I$, and M is a projection only if the range of P lies in the support; our fixture violates idempotence by 0.085 when it does not. The map also annihilates any off-support component of new rows, which is invisible at fit time and appears exactly in transport. H is appended unchanged, and because M acts on z alone, A ’s inference needs no H_B input. The moment must be computed on the declared valid-role masks with a *globally* centred channel: per-role re-centring drives the between-role mean shift to zero by construction, so a moment computed that way is already zero and erasing it erases nothing.

Why the coalition basis can differ from the local one—and when it cannot. The targets combine a service-view basis with cross-fitted sensitive residuals. Observing H_{AB} rather than H_A enlarges the conditioning set, which changes the residual and adds interaction columns; duplicating labels changes neither. In a fixture where $S = b_{H_B} \oplus b_Z$, the A -view interacting basis sees a cross-moment of 0.002 and the AB -view basis 0.124; duplicated H_A columns return exactly the same value as the originals. Cross-fitting controls one

bias; it is not evidence that the nuisance model is right, nor of conditional independence.

Aliases and information. Scaling the local matrix by $c > 0$ scales every eigenvalue by c , preserving order *and ties*, so the top- k projector is identical (difference exactly 0 in our fixture, including at a tied boundary). A mass-doubled local *hard* projection is therefore not a distinct method and must not be counted as a comparator. The alias fails under an absolute-tolerance rank rule (the same tolerance selects rank 2 at $c = 1$ and 3 at $c = 2$) and under ridge shrinkage. The local-*expanded* control changes the feature family, so it is a genuine, if imperfect, capacity control for the richer coalition basis. An invertible shrinkage cannot reduce information available to unrestricted attackers; a hard projection can. Neither fact guarantees improvement against a finite attacker slate. A full-span projection annihilates the specified empirical moments; a top- k one generally does not (residual 1.73 in our fixture). The trace criterion supports optimality of the retained subspace for that trace on the fitted moments—not minimal leakage and not a population statement.

The convex-correction statement, narrowed. With a fixed service-only logit offset ℓ_0 and a correction linear in a fixed basis Φ , empirical cross-entropy is convex in the correction parameters and its gradient at zero is $\Phi^\top(p_0 - Y)/n$ —exactly the empirical cross-moment the projection annihilates. Zero gradient plus convexity gives empirical optimality of the zero correction *for that family*; a direct search over the family in our fixture finds improvement exactly 0. The statement is false if the family contains an intercept and ℓ_0 was fitted on a disjoint fold, as the p_0 /mapper/monitor split does: the intercept gradient is the in-sample calibration error (0.026 in the smallest counterexample), and shifting it lowers cross-entropy by 0.0017. It does not carry over to a freely retrained H baseline, to arbitrary nonlinear corrections, or to population leakage. And zero finite moments do not imply nothing is recoverable: with $Z \mid S=1$ uniform on ± 1 and $Z \mid S=0 \sim N(0, 1)$, all first and second cross-moments vanish and $\mathbb{K}\{|Z| = 1\}$ recovers S with accuracy 1.0.

Relation to LEACE and earlier spectral PCRL. LEACE’s closed form is $r(x) = x - W^+ P_{W\Sigma_{XZ}} W(x - \mathbb{E}X)$ with $W = (\Sigma^+)^{1/2}$ [Belrose et al., 2023, Thm. 4.2–4.3]. When Σ has full rank and $P = I - P_{W\Sigma_{XZ}}$, the proposed map *is* this form with a different cross-moment substituted (our fixture agrees to 5×10^{-15}). When Σ is rank deficient they differ (by 2.28 in the fixture): LEACE’s subtractive form passes off-support components of new rows through unchanged, whereas projecting in whitened space deletes them. The specification must choose, and LEACE’s guarantees are proved for the subtractive form. Either way the construction has LEACE’s structure; its potentially useful distinction is *which* cross-moment—one conditioned on the coalition’s service view—and its application to strong existing channels with explicit rank controls. That is an adaptation, and we will describe it as one unless a full prior-work comparison supports more. No arbitrary-nonlinear privacy theorem is required for it to be worth measuring; what is refused is any reading of a low measured recovery as independence.

I Corrections carried forward from v3

1. **“LEACE dominates the repaired mechanism with no statistically detectable residence cost.”** The recovery half holds. The utility half does not: the difference is unresolved with an interval admitting 0.010 nats, it contains SPLINCE’s significant cost, and against the channel LEACE transformed the cost is significant at +0.0098. *Dominates* is withdrawn.
2. **“LEACE, SPLINCE and OptNet-C1 are statistically indistinguishable from J .”** Narrowed to unresolved differences. No equivalence test was run and all three point estimates lean towards J on local sex.
3. **J is not Pareto-dominant.** It recovers less and supplies less residence capability than several spectral arms, significantly so against riv16-L1. It is a trade.
4. **“Coordinate dependence was acting as an implicit regulariser.”** Narrowed to a hypothesis. Five ingredients changed together and no ablation separates them.
5. **The 63% rotation share** is an attained diagnostic value, not a supremum and not a causal decomposition of training benefit.

6. **“Zero population support”** for a race category is a *fitting-pool* count (1, 1, 0 by seed). A category missing from one pool is not absent from the population.
7. **The two carried numerical corrections are kept:** the `n1r8-C1` versus J local-sex deficit is +0.01210 person-weighted, not +0.0098; and “no endpoint significantly worse” for the nonlinear-penalty factor stays withdrawn, the counterexample being +0.00862 [+0.00168, +0.01556] unweighted.
8. **“Terminal 1’s experiment does not exist”** was stale. Study 3 is complete, committed and integrated here at 73903b7f.

J Superseded: the v3 pending note for Study 4

Revision v3 carried Study 4 as pending with ten preconditions, all satisfied at registration. Study 4 is now complete and integrated in §9; the v3 text is preserved unchanged at commit `ce5ba24a` on the v3 manuscript branch. The pending item is now Study 5 (Table 7); its integration checklist is `PENDING_INTEGRATION.md`.

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